Momentum analysis: put it together

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Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a model to compare a deflection option and its limits.

SUCCESS CRITERIA - I can...

1. Compute the required change before evaluating an option.
2. Keep impulse, velocity change and displacement distinct.
3. Communicate assumptions with the numerical result.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “deflection” and “lead time” to compare your model or result. Sentence starters are optional.

deflection: a change in the direction or path of motion

lead time: time available before the modeled event

system: the object or objects chosen for analysis

01 Bring back an earlier idea

Total momentum initially is (15, 0) kg·m/s. One fragment leaves with (3, 5) kg·m/s.
Find the other momentum vector and magnitude.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: shift = Δv × time; J = MΔv.
Fictional straight-line planning model. M is target mass; J is required impulse. This is not a real encounter forecast.

Worked reasoning: A 1 kg cart at +6 m/s sticks to a 2 kg cart at rest. Final velocity is +2 m/s. The moving cart changes momentum by -4 and the other by +4 kg·m/s.

Your turn: Draw both momentum changes and show how their sum matches the system change.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Represent and interpret

Organize this chain with units: impulse → velocity change → sideways shift. Which step uses target mass?

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: ① ② ③

04 Solve and justify

Target mass 2×10^{9} kg; desired shift 20 km over 4×10^{6} s.
Find requiredΔv and impulse.
Compare an option delivering 8×10^{6} N·s.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

If the target mass estimate is too low, how would a fixed delivered impulse's velocity change compare with the prediction?

Word help: estimate = find an approximate value

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: 1 2 3

06 Choose a method

A 4 kg cart moving right at 6 m/s stops in 3 s. Take right as positive.
Find signed impulse and average force.

Word help: impulse = net force acting over time; change in momentum

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

07 Exit check

Keep impulse fixed but double both target mass and lead time. What happens to modeled shift? Justify with an equation.

Word help: justify = support your answer with a reason or evidence

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A 2 kg cart at +4 m/s and a 1 kg cart at -2 m/s stick while an external impulse of -3 N·s acts.
Find final velocity and explain why setting initial momentum equal to final would fail.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3